

Sri Venkateswara College of Engineering, Sriperumbudur
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Electrical and Electronics Engineering

EE22811 Project Work – Second Review

Project Type: In-house/Industry

Batch No.:15

Name of the Internal Guide: Mr.N.Suresh

Name of the students with Roll No.

(1) Sanjay Kumar V - 2127220601063

(2) Shrimathi G - 2127220601070

Technical Area of the Project: Power Systems and Machine learning

Title of the Project : Power Quality Disturbance Classification Using Machine Learning.

Abstract:

Power quality disturbances (PQDs) such as voltage sag, swell, harmonics, and composite disturbances significantly impact grid stability and equipment lifespan, especially with increasing penetration of renewable energy sources. Conventional signal processing techniques, including Fast Fourier Transform (FFT) and Root Mean Square (RMS) analysis, exhibit limitations in detecting non-stationary and transient disturbances with low latency. This paper proposes a real-time PQD classification framework based on a Random Forest (RF) classifier optimized for edge deployment.

The system is trained on the XPQRS dataset comprising approximately 17,000 voltage signals across 17 disturbance classes. A total of 36 discriminative features are extracted from each signal, combining time-domain (RMS, crest factor), frequency-domain (Total Harmonic Distortion, spectral energy distribution), and wavelet-based features to capture both steady-state and transient characteristics. The dataset is split into 80% training and 20% testing, with feature normalization performed using StandardScaler and model validation conducted via 5-fold cross-validation. The RF model, consisting of 100 decision trees, performs multi-class classification through majority voting.

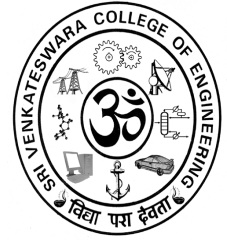
The proposed approach achieves a classification accuracy exceeding 95% with an inference time below 100 ms, making it suitable for real-time applications. Unlike deep learning-based methods requiring high computational resources (>2 GB GPU memory), the proposed system is lightweight and deployable on edge devices such as Raspberry Pi, reducing latency and power consumption. Additionally, an IoT-based architecture enables high-frequency (10 kHz) signal acquisition and real-time monitoring via a Node-RED dashboard.

The results demonstrate that the proposed framework provides an efficient, scalable, and computationally feasible solution for real-time PQD classification, ensuring improved grid reliability and enabling intelligent power quality monitoring in smart grid environments.

Signature of Internal Guide

Signature of the Students

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AY 2025-26 EE22811 Project Work Second Review Progress Sheet
(11/02/2026 to 18/03/2026)



Batch No:15

Section: B

Students Name & Roll No: Shrimathi G (70) and Sanjay Kumar V (63)

Title of the project:

The above listed students have reported to me on 20-03-2026 to finalize the project abstract and to review the power point slides presentation & simulation results. I hereby approve the same.

Date (project hours)	Absentees Roll No	Work done	Guide Approval with remarks
20/02/2026		Conducted a comprehensive study on power quality disturbances in electrical grids, focusing on voltage sag, swell, harmonics, flicker, and transient disturbances, along with their impact on system performance and equipment reliability.	
24/02/2026		Performed an extensive literature survey on conventional and advanced PQD detection techniques, including FFT, RMS, wavelet transforms, and machine learning-based classification methods.	
28/02/2026		Studied mathematical modeling and simulation of PQ disturbances based on IEEE standards to generate synthetic datasets for analysis.	
02/02/2026		Explored and compared multiple machine learning models including Random Forest, XGBoost, k-NN, and LightGBM for PQD classification.	
07/03/2026		Performed dataset preprocessing, including normalization, feature scaling, and splitting into training and testing datasets.	
09/03/2026		Designed system architecture and workflow covering signal input, feature extraction, classification, and output visualization.	
13/03/2026		Implemented the initial Random Forest classifier and trained the model using multiple decision trees with cross-validation and performed hyperparameter tuning as it performed better out of all the models.	
15/03/2026		Optimized the trained model to reduce overfitting and improve generalization using cross-validation and parameter adjustments.	
17/03/2026		Finalized the optimized Random Forest model and validated its performance for real-time PQD classification requirements.	
19/03/2026		Reviewed overall system performance and improvements with project guide and incorporated final suggestions for model refinement and deployment readiness.	

Date:

Place:

Signature of Project Guide

